Project4: Final Report

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Introduction

Linear regression is widely used to model the relationship between an outcome and a set of predictors. In moderate- to high-dimensional settings, variable selection becomes critical for accurately quantifying these relationships. Variable selection refers to identifying a subset of relevant predictors to include in a statistical model. Including irrelevant variables can increase variance, lead to overfitting, and reduce interpretability, while excluding important variables can introduce bias and result in incorrect inference.

A variety of variable selection methods have been developed, differing in both their procedures and performance. Classical approaches, such as backward selection, iteratively remove variables from a full model based on criteria such as p-values, Akaike Information Criterion (AIC), or Bayesian Information Criterion (BIC). In contrast, regularization methods such as lasso and elastic net incorporate penalties to control model complexity. Lasso applies an ℓ_1 penalty that can shrink coefficients exactly to zero, while elastic net combines ℓ_1 and ℓ_2 penalties to better handle correlated predictors. The performance of these methods depends on factors such as sample size, correlation among predictors, and signal strength.

In this simulation study, we compare several variable selection techniques in linear regression under controlled settings. Because the true model is known, simulation allows direct evaluation of each method's ability to identify relevant predictors, avoid selecting irrelevant variables, and accurately estimate regression coefficients. We consider settings with 20 predictors, of which 5 have true associations with the outcome, and evaluate performance across different sample sizes and correlation structures.

Methods

We conducted a simulation study to evaluate variable selection methods in linear regression. Analyses were implemented in R using the hdrm, glmnet, and stats packages, with parallel computation via furrr and future to improve computational efficiency.

Data were generated from a Guassian linear model $Y = X\beta + \epsilon$, where $X \in \mathbb{R}^{n\times 20}$ and $\epsilon \sim N(0, \sigma^2)$. The coefficient vector β contained five nonzero values ($X1 = 0.5/3, X2 = 1/3, X3 = 1.5/3, X4 = 2/3, X5 = 2.5/3$), while the remaining fifteen coefficients were set to zero. Predictors were generated using hdrm::gen_data() with an exchangeable correlation structure with three correlation settings ($\rho = 0, 0.35, 0.7$) considered to represent independent, moderately correlated, and highly correlated predictors. Two sample sizes (n = 250, 500) were considered, resulting in six simulation scenarios. For each scenario, 10,000 datasets were generated to ensure stable estimation of performance metrics, particularly for quantities such as type I error and coverage probabilities, which can exhibit substantial variability in smaller simulation studies.

We compared backward selection based on p-values, AIC, and BIC, along with lasso ($\alpha = 1$), and elastic net ($\alpha = 0.5$). For p-value-based backward selection, variables with p-values exceeding 0.10 were sequentially removed. This threshold was chosen to fall between the approximate p-value equivalents of AIC (~ 0.157) and BIC (~0.05), providing a compromise between less and more conservative selection criteria. AIC- and BIC-based methods used stepwise procedures to minimize their respective criteria utilizing the step function. Penalized regression methods were fit using glmnet, with tuning parameters selected via 10-fold cross-validation using both (λ_{min}) and (λ_{1se}). All models were fit without an intercept to align with the data generating mechanism in hdrm, where predictors are centered at zero. Including an intercept in this setting can introduce redundancy and affect variable selection behavior.

After variable selection, models were refit using ordinary least squares with selected predictors. Coefficient estimates, p-values, and 95% confidence intervals were extracted. To avoid conditioning performance metrics on selection, all predictors were retained in the simulation output for every replication, with estimates for unselected variables recorded as missing.

Selection performance was assessed using both aggregate and variable specific metrics. Aggregate performance was assessed using true positive rates (TPR) and false positive rates (FPR), defined as the proportion of true and noise predictors, respectively, selected within each replication and averaged across replications. Variable specific selection proportions were also calculated for each true predictor individually, while noise predictors were summarized collectively. Inference performance was evaluated using bias, empirical 95% confidence interval coverage, and error rates. Bias was defined as the average difference between estimated and true coefficients among selected true predictors, and coverage was defined as the proportion of 95% confidence intervals that contained the true parameter value. Type II error was evaluated separately for each true predictor and defined as the proportion of replications in which a predictor failed to be both selected and statistically significant. Type I error was defined using a family-wise approach as the proportion of replications in which at least one noise predictor (X6-X20) was both selected and statistically significant. This definition was chosen to better reflect the overall probability of making at least one false discovery within a fitted model rather than evaluating discoveries on a per-variable basis.

All simulation procedures were implemented using custom functions to generate data, perform variable selection, and extract model outputs. A fixed random seed was used to ensure reproducibility. Parallel computation was used to reduce runtime while maintaining reproducibility through controlled seeding.

Results

We evaluated seven variable selection methods across six simulation scenarios defined by sample size (n = 250, 500) and predictor correlation ($\rho = 0, 0.35, 0.7$). Aggregate selection performance is summarized in Figure 1, while Tables 1-3 present variable-level error rates, inference properties, and selection proportions.

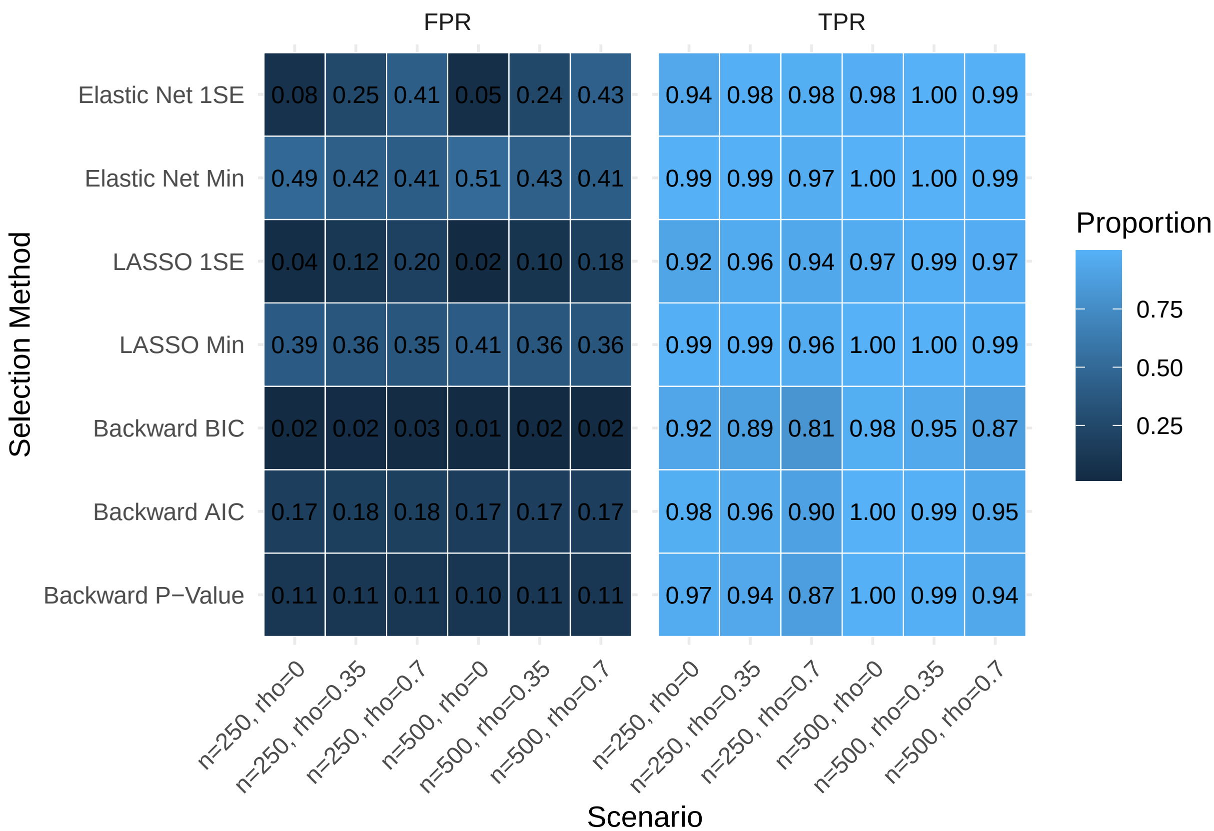

Figure 1: True Positive and False Positive Selection Rates Heat Maps

Figure 1 demonstrates a clear tradeoff between TPR and FPR across methods. Penalized methods using $\lambda_{\min}$ (LASSO Min and Elastic Net Min) achieved near perfect detection of true predictors (TPR ≈ 1.00), but at the expense of substantially higher false positive rates (FPR $\approx 0.35 - 0.51$). Comparatively, BIC consistently produced the lowest FPR (typically < 0.03), though with reduced TPR, particularly in more settings with higher correlation. AIC- and p-value-based methods showed intermediate performance, balancing moderate TPR with moderate FPR. Elastic net with the one standard error rule (λ_{1SE}) also demonstrated a more balanced performance relative to its λ_{min} counterpart, reducing false positives while maintaining high detection rates. Increasing predictor correlation generally increased FPR and reduced detection performance, while larger sample sizes modestly improved performance across methods.

Table 1 presents variable specific selection proportions. Predictors with stronger signals (X3-X5) were selected almost universally across methods and scenarios, whereas selection of X1, which had the weakest signal, varied substantially. BIC demonstrated the lowest selection proportion for X1, particularly under high correlation, while penalized methods using λ_{min} maintained substantially higher selection rates. Selection proportions for noise predictors reflected the FPR patterns observed in Figure 1, with penalized methods generally selecting more noise variables than conservative stepwise approaches.

Table 2 summarizes type II error rates for true predictors and type I error rates for noise predictors. Type II error was concentrated around covariates with weak signals particularly X1, while X2-X5 were almost always correctly identified (type II error near zero). Consistent with its conservative behavior, BIC exhibited higher type II error rates, whereas LASSO Min and Elastic Net Min achieved lower type II error rates. Family-wise type I error rates differed substantially across methods. Conservative approaches such as BIC maintained lower type I error rates, while penalized methods using λ_{min} frequently identified at least one noise predictor as statistically significant within a replication. Correlation among predictors generally increased type I error, while more conservative tuning choices using λ_{1SE} reduced false discoveries.

Table 3 summarizes bias and empirical 95% confidence interval coverage among selected true predictors. Bias was generally small in magnitude across methods and scenarios. Penalized regression methods showed slight negative bias consistent with shrinkage toward zero, while backward selection methods occasionally showed small positive bias. Coverage probabilities were generally close to the nominal 95% level for moderate correlation settings, but

Table 1: Selection proportions for true predictors (X1–X5) and pooled noise predictors (X6–X20) across simulation scenarios.

		$n = 250$						$n = 500$					
ρ	Method	X_1	X_2	X_3	X_4	X_5	X_6–X_{20}	X_1	X_2	X_3	X_4	X_5	X_6–X_{20}
$\rho = 0$	Backward P-Value	0.829	1.000	1.000	1	1	0.105	0.976	1.000	1	1	1	0.103
	Backward AIC	0.881	1.000	1.000	1	1	0.172	0.987	1.000	1	1	1	0.166
	Backward BIC	0.616	0.998	1.000	1	1	0.022	0.882	1.000	1	1	1	0.014
	LASSO Min	0.944	1.000	1.000	1	1	0.394	0.995	1.000	1	1	1	0.405
	LASSO 1SE	0.596	0.993	1.000	1	1	0.038	0.826	1.000	1	1	1	0.020
	Elastic Net Min	0.964	1.000	1.000	1	1	0.491	0.998	1.000	1	1	1	0.505
	Elastic Net 1SE	0.712	0.998	1.000	1	1	0.076	0.912	1.000	1	1	1	0.050
$\rho = 0.35$	Backward P-Value	0.710	0.996	1.000	1	1	0.112	0.926	1.000	1	1	1	0.108
	Backward AIC	0.784	0.998	1.000	1	1	0.180	0.952	1.000	1	1	1	0.171
	Backward BIC	0.470	0.984	1.000	1	1	0.024	0.757	1.000	1	1	1	0.015
	LASSO Min	0.937	1.000	1.000	1	1	0.356	0.993	1.000	1	1	1	0.361
	LASSO 1SE	0.801	0.999	1.000	1	1	0.119	0.944	1.000	1	1	1	0.096
	Elastic Net Min	0.953	1.000	1.000	1	1	0.423	0.997	1.000	1	1	1	0.432
	Elastic Net 1SE	0.915	1.000	1.000	1	1	0.248	0.989	1.000	1	1	1	0.240
$\rho = 0.7$	Backward P-Value	0.452	0.911	0.998	1	1	0.114	0.686	0.996	1	1	1	0.111
	Backward AIC	0.548	0.940	0.999	1	1	0.182	0.756	0.997	1	1	1	0.172
	Backward BIC	0.240	0.804	0.991	1	1	0.028	0.396	0.974	1	1	1	0.018
	LASSO Min	0.826	0.991	1.000	1	1	0.353	0.939	1.000	1	1	1	0.356
	LASSO 1SE	0.715	0.978	1.000	1	1	0.196	0.871	0.999	1	1	1	0.183
	Elastic Net Min	0.858	0.994	1.000	1	1	0.407	0.954	1.000	1	1	1	0.413
	Elastic Net 1SE	0.888	0.996	1.000	1	1	0.413	0.971	1.000	1	1	1	0.434

deviations were observed under high correlation ($\rho = 0.7$), in some cases falling below 0.93. Larger sample sizes modestly improved coverage, although nominal levels were not consistently achieved. Overall, no single method uniformly outperformed others. Methods that prioritized detection of true predictors, such as LASSO Min and Elastic Net Min, obtained higher false positive rates and shrinkage bias, whereas more conservative approaches reduced false discoveries at the expense of missing weaker signals.

Table 2: Type II error (X1–X5) and Type I error (X6–X20) across scenarios.

		$n = 250$						$n = 500$					
ρ	Method	X_1	X_2	X_3	X_4	X_5	X_6–X_{20}	X_1	X_2	X_3	X_4	X_5	X_6–X_{20}
$\rho = 0$	Backward P-Value	0.257	0.000	0.000	0.000	0	0.560	0.045	0.000	0	0	0	0.552
	Backward AIC	0.260	0.000	0.000	0.000	0	0.568	0.046	0.000	0	0	0	0.556
	Backward BIC	0.384	0.002	0.000	0.000	0	0.272	0.118	0.000	0	0	0	0.184
	LASSO Min	0.274	0.001	0.000	0.000	0	0.548	0.048	0.000	0	0	0	0.549
	LASSO 1SE	0.414	0.007	0.000	0.000	0	0.316	0.175	0.000	0	0	0	0.211
	Elastic Net Min	0.275	0.001	0.000	0.000	0	0.548	0.048	0.000	0	0	0	0.548
	Elastic Net 1SE	0.329	0.002	0.000	0.000	0	0.440	0.092	0.000	0	0	0	0.384
$\rho = 0.35$	Backward P-Value	0.389	0.007	0.000	0.000	0	0.603	0.121	0.000	0	0	0	0.592
	Backward AIC	0.394	0.009	0.000	0.000	0	0.608	0.124	0.000	0	0	0	0.597
	Backward BIC	0.529	0.016	0.000	0.000	0	0.292	0.243	0.000	0	0	0	0.200
	LASSO Min	0.534	0.021	0.000	0.000	0	0.447	0.210	0.000	0	0	0	0.459
	LASSO 1SE	0.531	0.020	0.000	0.000	0	0.178	0.193	0.000	0	0	0	0.194
	Elastic Net Min	0.531	0.021	0.000	0.000	0	0.499	0.207	0.000	0	0	0	0.514
	Elastic Net 1SE	0.587	0.025	0.000	0.000	0	0.129	0.241	0.000	0	0	0	0.132
$\rho = 0.7$	Backward P-Value	0.648	0.134	0.006	0.000	0	0.627	0.411	0.010	0	0	0	0.613
	Backward AIC	0.654	0.142	0.007	0.000	0	0.633	0.417	0.012	0	0	0	0.610
	Backward BIC	0.760	0.196	0.010	0.000	0	0.332	0.604	0.026	0	0	0	0.238
	LASSO Min	0.805	0.284	0.022	0.001	0	0.351	0.594	0.035	0	0	0	0.363
	LASSO 1SE	0.831	0.292	0.023	0.001	0	0.126	0.621	0.035	0	0	0	0.128
	Elastic Net Min	0.804	0.283	0.022	0.000	0	0.391	0.589	0.034	0	0	0	0.419
	Elastic Net 1SE	0.850	0.320	0.026	0.001	0	0.105	0.649	0.041	0	0	0	0.107

Discussion

This simulation study compared several commonly used variable selection methods under varying sample sizes and predictor correlation structures. Overall, the results demonstrated a clear tradeoff between aggressive variable detection and control of false discoveries. Penlaized regression methods using λ_{min}, particularly LASSO Min and Elastic Net Min, achieved the highest true positive rates and lowest type II error rates, even in highly correlated settings. However, this improved sensitivity came at the cost of substantially higher false positive and type I error rates, indicating a greater tendency to include irrelevant predictors. In contrast, BIC-based backward selection was considerably more conservative, producing the lowest false positive rates and strongest control of type I error, but frequently failing to detect weaker

Table 3: Bias and empirical 95% confidence interval coverage among selected true predictors.

			X_1		X_2		X_3		X_4		X_5	
ρ	n	Method	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage
$\rho = 0$	$n = 250$	Backward P-Value	0.020	0.965	0.000	0.946	0.000	0.948	0.000	0.943	0.000	0.947
		Backward AIC	0.015	0.967	0.000	0.945	0.000	0.948	0.000	0.942	0.000	0.945
		Backward BIC	0.040	0.956	0.000	0.952	0.000	0.950	0.000	0.946	0.000	0.949
		LASSO Min	0.006	0.966	-0.002	0.942	-0.002	0.946	-0.001	0.943	-0.001	0.945
		LASSO 1SE	0.038	0.957	-0.001	0.953	-0.002	0.949	-0.001	0.943	-0.001	0.949
		Elastic Net Min	0.004	0.963	-0.002	0.943	-0.002	0.947	-0.002	0.943	-0.002	0.947
		Elastic Net 1SE	0.027	0.964	-0.002	0.948	-0.003	0.949	-0.002	0.944	-0.002	0.948
	$n = 500$	Backward P-Value	0.003	0.965	0.000	0.951	0.000	0.949	0.001	0.947	0.000	0.946
		Backward AIC	0.002	0.956	0.000	0.951	0.000	0.949	0.001	0.947	0.000	0.945
		Backward BIC	0.010	0.970	0.000	0.953	0.000	0.953	0.001	0.949	0.000	0.949
		LASSO Min	0.000	0.949	-0.001	0.952	0.000	0.949	0.000	0.946	0.000	0.945
		LASSO 1SE	0.013	0.968	0.000	0.952	0.000	0.953	0.000	0.949	0.000	0.948
		Elastic Net Min	0.000	0.946	0.000	0.953	0.000	0.950	0.000	0.946	0.000	0.946
		Elastic Net 1SE	0.007	0.969	-0.001	0.952	-0.001	0.952	0.000	0.949	0.000	0.947
$\rho = 0.35$	$n = 250$	Backward P-Value	0.037	0.956	0.002	0.943	0.002	0.929	0.003	0.935	0.002	0.937
		Backward AIC	0.029	0.960	0.000	0.943	0.001	0.933	0.002	0.933	0.001	0.938
		Backward BIC	0.063	0.944	0.010	0.949	0.008	0.931	0.009	0.936	0.008	0.938
		LASSO Min	-0.014	0.978	-0.025	0.933	-0.024	0.924	-0.023	0.927	-0.024	0.929
		LASSO 1SE	-0.001	0.986	-0.026	0.932	-0.025	0.926	-0.024	0.930	-0.025	0.930
		Elastic Net Min	-0.014	0.972	-0.024	0.937	-0.023	0.927	-0.022	0.932	-0.023	0.932
		Elastic Net 1SE	-0.022	0.983	-0.036	0.930	-0.035	0.923	-0.034	0.926	-0.035	0.925
	$n = 500$	Backward P-Value	0.008	0.970	0.000	0.941	0.001	0.942	0.001	0.941	0.000	0.948
		Backward AIC	0.005	0.970	0.000	0.940	0.001	0.941	0.000	0.941	0.000	0.948
		Backward BIC	0.021	0.966	0.003	0.942	0.004	0.942	0.004	0.943	0.003	0.947
		LASSO Min	-0.016	0.936	-0.017	0.930	-0.016	0.931	-0.016	0.935	-0.017	0.939
		LASSO 1SE	-0.011	0.974	-0.016	0.933	-0.015	0.934	-0.016	0.936	-0.016	0.938
		Elastic Net Min	-0.015	0.935	-0.015	0.933	-0.015	0.934	-0.015	0.936	-0.016	0.942
		Elastic Net 1SE	-0.023	0.935	-0.024	0.925	-0.024	0.927	-0.024	0.930	-0.025	0.931
$\rho = 0.7$	$n = 250$	Backward P-Value	0.098	0.930	0.025	0.957	0.007	0.933	0.005	0.929	0.006	0.929
		Backward AIC	0.081	0.939	0.018	0.960	0.004	0.934	0.003	0.930	0.004	0.931
		Backward BIC	0.148	0.851	0.054	0.943	0.024	0.929	0.021	0.920	0.022	0.920
		LASSO Min	-0.012	0.985	-0.045	0.927	-0.047	0.925	-0.048	0.924	-0.047	0.926
		LASSO 1SE	-0.006	0.992	-0.051	0.933	-0.057	0.919	-0.058	0.919	-0.057	0.921
		Elastic Net Min	-0.016	0.984	-0.043	0.927	-0.045	0.926	-0.046	0.927	-0.045	0.927
		Elastic Net 1SE	-0.036	0.993	-0.058	0.925	-0.059	0.927	-0.059	0.928	-0.058	0.931
	$n = 500$	Backward P-Value	0.040	0.955	0.004	0.937	0.002	0.939	0.003	0.938	0.003	0.933
		Backward AIC	0.032	0.960	0.002	0.936	0.000	0.941	0.002	0.939	0.002	0.937
		Backward BIC	0.074	0.928	0.018	0.946	0.012	0.930	0.014	0.926	0.014	0.927
		LASSO Min	-0.023	0.977	-0.032	0.923	-0.033	0.923	-0.032	0.927	-0.032	0.923
		LASSO 1SE	-0.021	0.992	-0.038	0.919	-0.040	0.915	-0.038	0.920	-0.038	0.918
		Elastic Net Min	-0.023	0.964	-0.030	0.925	-0.031	0.924	-0.029	0.929	-0.030	0.926
		Elastic Net 1SE	-0.036	0.949	-0.040	0.924	-0.042	0.923	-0.040	0.928	-0.040	0.926

signals such as X1.

These findings highlight that no single variable selection approach is universally optimal and that the preferred method depends on the primary goal of the anlaysis. Less conservative penalized apporaches may be preferable when maximizing detection of potentially relevant predictors is most important, whereas more conservative approaches such as BIC may be preferable when minimizing false discoveries or obtaining a more parisomnious model is the primary objective. Elastic Net 1SE demonstrated a more balanced tradeoff between sensitivity and specificity. Predictor correlation substantially affected performance across methods. As correlation increased, false positive rates and type I error generally increased while detection of weaker signals became more difficult. Larger sample sizes modestly improved performance, though they did not fully eliminate the tradeoff between identifying true predictors and controlling false discoveries. More broadly, these results emphasize the importance of collaboration between statisticians and subject matter experts when determining which predictors are most important to retain in a final model.

Several limitations should be considered. The simulation scenarios were limited to Gaussian linear models with an exchangeable correlation structure and a fixed number of predictors. Performance may differ in higher-dimensional settings, alternative correlation structures, or non-Gaussian outcomes. Additionally, post-selection inference was performed by refitting ordinary least squares models without adjusting for selection uncertainty, which may have contributed to the observed deviations from nominal confidence interval coverage. Finally, only a limited set of tuning parameter choices and selection procedures were considered. Overall, this simulation study demonstrates the important balance between variable detection, false dicovery control, and inference quality. These findings suggest that the choice of variable selection procedure should be guided by the scientific goals of the analysis rather than relying on a single method in all settings.

Reproducible Research Information

https://github.com/eromo96/BIOS6624.git

Simulation Functions Location: /Users/eduardo/Eduardo/UColorado PhD Bios/Spring 26/BIOS6624 Advanced Statistical Methods/BIOS6624/Project4/Code